CHAPTER 4

Agriculture and Allied Activities

1. AGRICULTURE

Agriculture is one of the major economic activities of the people in the State, contributing about 6% to the State Domestic Product (SDP). However, due to high cost and non availability of labour the growth has been reduced in recent years. Extension activities are being further strengthened by regular coverage of trainings and exposures of farmers to advances in agricultural and need based technical support through Krishi Vigyan Kendras and agricultural infrastructure. The activities of the directorate are directed to arrest the deceleration of growth in the agriculture sector by providing assistance towards purchase of machinery to reduce the cost of production, increase productivity by improvement in irrigation, promoting dug wells, rainwater harvesting, desilting of water bodies, use of quality seeds including hybrids, optimal use of manures and fertilizers.

a. Some of the significant initiatives taken to improve production and productivity and reduction of cost cultivation etc. are as under:

- A subsidy of 17% for cultivation of flowers from State fund besides the subsidy of Government of India.
- Increase in incentive for sale of paddy from Rs. 500/- per m.t. to Rs.1000/- per m.t.
- Additional 25% State subsidy for purchase of power tillers in addition to central subsidy.
- Subsidy for laying of irrigation pipelines.
- Increase in subsidy upto 90% for 3 HP & 50% upto 10 HP for purchase of water pumps.
- Increase in support price to Arecanut to ensure price of Rs.100/- per kg. of arecanut.
- State subsidy of Rs. 5.00 lakh in addition to central subsidy of Rs. 5.00 lakh limiting to 50% of actual cost to a group of farmers who produce a certificate from the designated societies in the State that organic manure is used for agricultural produce.
- Enhancement of subsidy to 50% of cost with cost ceiling of Rs. 25,000/- per ha. for sprinkler irrigation system with net work of RPVC pipes.
- Enhancement of cost of 25% State subsidy with ceiling upto Rs. 40,000/- per ha. for HDPE pipe line net work.
- A subsidy of 25% up to a maximum of Rs.10,000/- per implement for purchase of implements like harvester,

transplanters, brush cutters, paddy threshers.

- Custom service facilities by providing machinery like tractors, power tillers & bulldozers to the paddy crop growers.
- Subsidizing 50% of insurance premium to small and marginal farmers for cover against accidental fire hazards to cashew & sugarcane crops.
- Waiving of outstanding loan availed during early 1980's from Agriculture Department by scheduled caste farmers.

b. Soil & Water Conservation

In order to protect 18,000 hectares of paddy lands locally named as khazan lands, from saline water inundation, construction of embankments, marginal to the rivers and creeks has been undertaken. Various programmes are under implementation for their repair and remodeling which are categorized as follows:

- Category I: Repairs to protective embankments by providing 50% subsidy subject to a maximum Rs. 6000/- per ha. of area benefited to Tenant's Association/ Beneficiaries on successful completion of works. Most of the protective bunds in the State are repaired and maintained under this scheme. During the year 2007-08, 38 notified bunds are undertaken for its repair. It is estimated that a subsidy amounting to Rs. 50.00 lakh will be paid on these works during this financial year.
- Category II: The repairs of protective embankment of lesser width are renovated with wider and structurally stronger sections with 100% initial investment by Government. The beneficiaries' contribution is only 10% of the cost incurred to be paid in ten annual installments with 6% simple interest. The Government's contribution is up to 90%. There is a State Level High Powered Committee to monitor the scheme. During the year 2007-08, 11 works are in progress and it is proposed to take up 3 more works. It is estimated that an amount of Rs. 150.00 lakh will be spent on the ongoing works.

c. Western Ghat Development Programme (WGDP)

Western Ghat Development Programme (WGDP) Scheme is implemented in three talukas of Sattari, Sanguem and Canacona. The aim of the scheme is to restore ecosystem and uplift the standard of living of people residing in Western Ghat region in sustainable manner. The Self Help Groups (SHGs) and Users Groups (UGs) are actively involved in taking up the conservation and production programmes, organising training for officers and farmers, exposure visits and Krishi Melas to motivate the farmers. During the year 2007-08, 4 trainings and 3 Krishi Melas were held and 900 farmers attended these trainings.

The activities/works for soil water conservation and rainwater harvesting are implemented under the scheme. The activities of crescent shape conservation trenches, terracing, bunding and leveling,

gully control by loose boulder check dams, nullah bank stabilizations are taken up for soil and water conservation. Work of farm pond/dugout ponds are taken for water harvesting. The achievement under land development is 2.4 ha. covering farmers under ST category.

Under the production programmes, household production systems are being undertaken and till date one programme each under general category and ST category have been taken up. The units promoted under this activity would include assistance for farm implements suited to location and Agro processing units like Cashew Juice extractors, Powdering machine for spices, Plate and saucer maker, Drier for agro-products and metal storage bins etc.

The plantation of improved varieties of grafts of rainfed fruit trees like Awla, Cocum, Tamarind etc, are taken up in wasteland areas by the SHG and by the individual farmers to increase the income. A total area of 50 ha. has been covered this year at an expenditure of Rs. 69764/- for farmers under general category and Rs.69079/- under scheduled tribe category. About 5203 saplings plants were supplied to 450 farmers.



A "Krishi Mela and Exhibition cum sale of hilly vegetables and fruits and tubers was held in Canacona and Sattari talukas. The Mela was held to reaffirm the Government support and create marketing opportunities for the Self Help Groups generating large quantities of produce from hilly areas of Sanguem, Canacona and Sattari talukas. 59 SHGs having approximately 1180 farmers participated. More than 24,000 people visited the Mela.

d. Revitalization of Ponds/Lakes

The work of Revitalisation of ponds/lakes is being undertaken with the involvement of User Groups, NGOs or private individuals. The assistance provided is 100% in case of UGs/NGOs as grant-in-aid and 50% in case of private individuals as subsidies.



e. National Watershed Development Programme for Rainfed Areas (NWDPRA)

The State of Goa had taken up 27 micro watersheds in four blocks of the two districts of the State with total project cost of Rs. 1254.00 lakh to develop an area of 22721 hectare. Out of the total treatable area, a total of 14221 ha. has been treated until March 2007(64.37%) at a cost of Rs.782.41 lakh (62.39%). 7 watersheds have been saturated by March 2007 and the remaining 20 have been given an extension period up to March 2008, for completion.

A total of 30 micro watersheds covering 4 talukas have been selected to be taken up under NWDPRA during the Eleventh Five Year Plan. The projected area to be treated is 26707 ha. at a total projected cost of Rs.16.02 crore. Allotment of 25 new watersheds have been undertaken by various Project Implementation Agencies (PIAs) and initial work of recruitment of team members, holding of Gram Sabhas and signing of MoU with District Nodal Agency is in process.

The allotment of watersheds in the South and North Goa districts are as follows:

SOUTH GOA

Sl. No	Name of PIA	Name of watersheds allotted
1	Don Bosco-The Quepem Salesian Society Premnagar	a) Gokuldem b) Padi c) Kiscond d) Ambaulim
2	Mineral Foundation of Goa, Vaglo Building, Panaji	a) Rivona b) Malkarnem c) Kavrem Male

NORTH GOA

Sl. No	Name of PIA	Name of watersheds allotted
1	The Energy Resources Institute, Dona Paula-Goa	a) Ladpe b) Dodamarg c) Kasarpal d) Upsa e) Advalpal
2	Gram Vikas Kendra, Savoiverem, Ponda-Goa	a) Priol b) Apewal c) Borim d) Carai
3	Nirmal Vishwa-Goa Sarjotisi Colony Khadpaband Ponda-Goa	a) Nirankal b) Torla c) Paj-Conshe d) Wubjem
4	Keshav Seva Sadhana Durga Nagar, Borim Ponda- Goa	a) Mapa b) Panchwadi c) Amlai
5	Mineral Foundation of Goa, Vaglo Building, Panaji	a) Mayem-Tikhajan b) Mayem-Haturli

f. Horticulture

The National Horticulture Mission is being implemented in the State to promote Horticulture & Floriculture in the farming community of the State so that their outlook changes from Sustenance Agriculture to Commercial Agriculture. Farmers are being encouraged and supported financially to take up cultivation of flowers, Cashew, Mango, Kokum, Awla, Spices, Medicinal & Aromatic crops. Quality planting material is also being made available from Government & private recognized nurseries outside the State and within the State. Assistance is provided by way of technical know-how and subsidies are given for setting up green houses, shade net and other commercial ventures in floriculture, mushroom cultivation, organic farming & organic certification. Assistance is also provided for Post Harvest facilities like setting up markets in towns & rural areas for value addition machinery, grading machines, weighing machines, cold storage etc. This assistance is provided to both, Public & Private Sector.

A market project in Sanquelim is approved for assistance to the tune of Rs 3.25 lakh. Organic certification project in the private sector is being considered for an amount of Rs. 15.00 lakh. During this year, 150 ha. have been brought under cashew cultivation with new elite varieties. 390 ha. old senile cashew plantation have been rejuvenated with elite varieties of cashew like V7, Balli 2, V4 & V8. Around 400 farmers are trained so far under National Horticulture Mission. 10 officers were also trained in the advance system of irrigation. To promote production of tender coconuts 10,000 dwarf coconut saplings were provided to various cultivators in the State.



g. Support Price for arecanut/coconut

Arecanut growers are ensured minimum price of Rs.100/- per kg of arecanut. Whenever there is a price fall below Rs. 100/- , the difference is paid by the Department of Agriculture as support price. An amount of Rs. 110.40 lakh was paid in the year 2006-07 for 2167 growers and in the year 2007-08 an amount of Rs. 125.00 lakh is proposed to be paid. A support price of Rs. 4/- per coconut is proposed during this year as the farmers income is thus safeguarded by the Government in case of fall in prices below uneconomical levels.

h. Promotion of Vegetable Crops

During the year 2007-08, the cultivation of vegetable is promoted by providing subsidy of 50% of cost on the seed. 800 kg. of seed is supplied to growers with subsidy of Rs. 1.50 lakh upto November 2007.

i. Departmental farm

The Directorate of Agriculture is having two farms i.e. at Codar and Kalay spread over an area of 150 ha. These farms are mainly utilized for multiplication of good quality planting material, maintenance of orchards, vegetable seed production etc. Genuine and quality planting material is the first and foremost priority of these farms. In order to produce good quality planting material, farms are maintaining progeny orchards under cashew varieties like Vengurla 4, 6 & 8 (35 ha), mango varieties like mancurad, musarad, ratna, amrapali, etc., (12 ha) & coconut (15 ha). Under National Horticulture Mission, the department has established infrastructure like shed net poly house on an area of about 2000 sq.mts. During the year 2007-08, the quality planting material produced on these farms are mango grafts (38,000), cashew grafts (95,500), coconut seedlings (84,500), vegetable seed production (780 kgs) and paddy seed about 13 quintals

j. Agricultural Implements/Machinery

Adequate machinery is required to bring more land under cultivation, to improve productivity and to reduce the cost of cultivation. The department provides custom

services to the farmers by providing tractors, power tillers and bulldozers and has a fleet of 44 tractors, 2 bulldozers and 4 power tillers. During the year, 17,770 hours of tractor, 367 hours of bulldozer and 300 hours of power tiller work has been carried out till November' 07. Under various departmental schemes, tractors, power tillers and brush/weed cutters are provided to the farmers on subsidy, which help the farmers to carry out their agricultural operation in their own field as well as hire them to the needy farmers from the villages. During the current year, 132 farmers have taken benefit of purchase of tractors, power tiller and weed cutters.

For popularizing agricultural machinery through Self Help Groups (SHGs), registered Non Government Organizations (NGOs), Farmer's Club, Farmer's Society etc. a subsidy of 70% is provided. This included 25% by Government of India & 45% by the State for purchase of agricultural machinery/equipment etc. Individual farmers are provided 50% subsidy towards the purchase of Power Tiller. During the year 2007-08, it is proposed to cover 500 farmers with assistance for mechanization and purchase of power driven equipments.

k. Agricultural extension activities

- During the year 2007-08, 3 students are deputed for higher education in Agriculture at the Agriculture University at Dapoli in Maharashtra.
- Four outstanding farmers under category of "Best Farmer", "Best Horticulturist" "Best Invocative Farmer", and "Best Farmer Adopting Diversification and Integrated Farming" were honoured with cash prize of Rs. 10,000/- each and a traditional lamp.
- The Directorate of Agriculture participated in ZAK SALAAM INDIA Expo 2007 at Singapore from 15th to 19th August 2007, to expose the agricultural potential of this State. A stall of 9 sq. mts. depicting coconut, cashew nut, other fruits like banana, pumello, spices, paddy, ragi, groundnut and their processed products were exhibited. A good number of inquiries for export of cashew nut, cashew feni and other agro-based products like pickles have been received.
- A State Level Krishi Mahotsav 2007 was organized in association with the Goa Bagayatdar Sahakari Kharedi Vikri Saunstha Maryadit, Ponda under the Central Government Programme of National Horticulture Mission from 18th to 21st January 2007 at the Sports Complex Ground, Ponda-Goa. In all, 130 entrepreneurs, Government Organizations etc. participated in the Mahotsav. 15,000 farmers and public in general visited the exhibition. An exclusive garden depicting various cropping patterns in agriculture practiced by farmers in Goa was laid out at the central place of exhibition.
- A Special programme was held at Farmers Training Centre, Ela-Old Goa and ICAR, Ela-Old Goa. A total of 1000 farmers participated in the programme

and were acquainted with the latest agricultural technologies. It has been noted that farmers from predominantly agriculture based talukas of Goa such as Sanguem, Canacona, Quepem and Sattari are not thoroughly exposed to various ongoing departmental programmes and new technological introduction in agriculture and allied facilities.

- Various competitions for vegetables, fruits, and fruit carving, flower arrangements, cooking etc. were held for the public during the exhibition. Training and live demonstrations for farmers were also conducted.
- Directorate of Agriculture has organized various exposure visits of farmers within and outside the State regularly. This has exposed the farmers to technological advancement at State Agriculture Management and Extension Training Institute (SAMETI) Ella, Old-Goa. Special attention has been paid to train SC/ST and women farmers and exclusive targets supported by funds are allotted to cover these farmers under various training programme. Farmers are also taken on tour to ICAR, Old-Goa to demonstrate integrated farming in Agriculture. In all 7,000 to 8,000 farmers have been trained during 2007-08.
- Transport facilities from villages to training sites are provided to farmers free of cost by the directorate. Need based training programmes, for a group of farmers have also been arranged in villages as well as at the SAMETI. Technical experts from Department of Agriculture, ICAR, line departments and technical institutions, etc. have been called in regularly to impart the trainings.
- Support of AIR, Doordarshan and other private TV channels is also taken to regularly broadcast agricultural and allied sector programmes, exclusively for farmers and public in general of the State. Toll free telephone facilities are regularly utilized by farmers to solve their individual agricultural problems.
- Facilities of lodging and boarding have been provided to trainee farmers at the SAMETI hostel at Old-Goa. Training kits are also distributed to farmers free of cost during the training. Live demonstrations have been organized in mushroom production, budding and grafting methods, advanced cultivations practices, various mechanical advancement in agriculture, etc. Farmers are also exposed to different package of practices in crop production including floriculture development.

I. Manures and Fertilizers

- **Soil Conditioners:** In view of the acidic nature of soils in the State, the directorate provides soil conditioners like Rock Phosphate to the farmers. The soil conditioners are made available at 75% subsidized rate with a maximum assistance of Rs.1500/- per ha., the recommended dose being 500

kgs/ha. The ceiling is 4 ha. per cultivator.

- **Organic Manure:** To improve the productivity level in the State, organic manure procured from Solid Waste Treatment Plant is provided to the farmer against a token amount of Rs.100/- per tonne. 500 tonnes of organic manure will be provided during the year.

m. Promotion of Cereal Crops

- To encourage farmers to use good quality seed and for seed replacement of old seed, 50% subsidy is made available at source to the farmers purchasing Certified paddy seed through registered Societies. During Kharif season 260 M.T. of paddy has been sold for which Rs.21.00 lakh have been disbursed as subsidy.
- To have disease free plant population a scheme of 100% seed treatment is operated by the directorate. The farmers have been provided 5.6 tonne of Trichoderma and about 1300 M.T. of seed has been treated so far.
- The department provides a support price of Rs. one per kg. of paddy when sold to Registered Societies and rice mills. Till date a support price of Rs.5.01 lakh has been paid.

n. Development of Sugarcane

- **Assistance for expansion of new area in sugarcane:** The local cane production is at stand still in recent years. The lone sugar factory in the State has to rely considerably on adjoining area of Karnataka to meet its sugarcane requirement. To boost the expansion of area under sugarcane, 25% subsidy limited to Rs. 6000/- ha. is provided towards purchase of required inputs such as seed, manure fertilizers, bio-pesticides, weedicides, pesticides etc. for cultivating new area (plant cane). The target of 100 ha. is earmarked for the State and the same will be achieved by March 2008.
- **Assistance for application of manures and fertilizers for ratoon crop:** Ratoon crops occupies the major area under sugarcane, which is ill fed and accounts for low productivity. This scheme aims at increasing sugarcane production, by providing financial assistance to ratoon crops by application of recommended dose of manure and fertilizers. 25% financial assistance limited to Rs. 4000/- ha. is provided under this scheme component. A target of 460 ha. is earmarked for the State and the same will be achieved by March 2008.
- **Financial assistance towards purchase of sugarcane seed:** Seed is one of the vital and costly inputs in sugarcane production. 25% financial assistance limited to Rs. 2500/- is provided towards purchase of quality sugarcane seed. A target of 200 ha. is earmarked for the State which will be achieved by March 2008.
- **Support price for Sugarcane:** Sugarcane support price is already paid

to 1237 growers @ Rs. 200/- per ton. The support price is paid for 56000 M.T. sugarcane produced locally amounting to Rs. 112.00 lakh.

o. Special Programme for SC & ST farmers

- **Assistance for "Jalkund"/Water storage:** Many SC/ST farmers are engaged in cultivation of vegetables like Cucumber, Bottle gourd, Ridge gourd, Pumpkin, Chillies, Ashgourds etc. alongwith traditional flowers on hill slopes during kharif season. Some of these vegetables are sown much before the monsoon season in order to obtain premium price for the early crop. The dry spells during monsoon sometimes demands for alternate provision for water to safeguard the crop. The present scheme provides 50% assistance to SC/ST farmers limited to Rs. 3000/- for Jalkund/Water Storage structure of size 2 m x 2 m with a depth of 1.5 m lined with thick plastic/silpaulin and provision of adequate fencing around it to prevent fall of animals. Pucca Jalkund/water storage structures with laterite stones and cement are provided with 50% subsidy limited to Rs. 8000/- having dimensions 2 m x 2 m x 1.5 m. A target of 105 Jalkunds is earmarked for the State during 2007-08. About 60% achievements are anticipated by March 2008.
- **Assistance for inputs purchased for agriculture:** Assistance of 50% is provided to SC & ST farmers for purchase of all the agricultural inputs, seeds, pesticides, fertilizers, tools & equipments.

p. Crop Insurance

The scheme is aimed to provide insurance coverage and financial support to farmers in case of failure of notified crops. Presently paddy, pulses, Ragi, groundnut and sugarcane crops are covered under the scheme. The loanee as well as non loanee farmer can avail the benefits of the scheme.

q. Rainfall

The total rainfall received during the calendar year 2007 was 3,592 mm. The rainfall during the year 2007 was above the average and there was heavy rainfall during critical period of crop growth for paddy and groundnut whereby the average yields are likely to be 20% and 50% below the normal for paddy and groundnut respectively. The trend in rainfall in the State from 1992 to 2007 is given in table 4.1.

Table 4.1 : Rainfall recorded from 1992 to 2007

Year	In. mm.
1992	2,778.2
1993	2,558.3
1994	2,894.4
1995	3,555.6
1996	2,880.9
1997	3,366.9
1998	3,078.9
1999	3,680.4
2000	3,511.6
2001	2,128.1
2002	2,270.4
2003	2,682.9
2004	2,156.0
2005	3,271.0
2006	2,644.1
2007	3,592.5

r. Crop Forecast

Table 4.2 : Crop forecast for major foodgrain crops for the year 2007-08

Sl. No.	Crop	Kharif 2007		Rabi 2008 (Anticipated)	
		Area (ha.)	Production Anticipated (tonne)	Area (ha.)	Production (tonne)
1	Rice	34261	65918	17350	76000
2	Pulses	569	400	11600	12700
3	Groundnut	550	500	2500	4750
	TOTAL	35380	66818	31450	93450

Due to unseasonal rains in the Kharif season there is a decline of 20% in Rice production and 50% in Groundnut production.

2. ANIMAL HUSBANDRY & VETERINARY SERVICES

The policies and strategies of Livestock development are aimed at improving the genetic material of indigenous cattle and buffaloes, development of Veterinary Health Cover, Poultry and Piggery Development, Feed and Fodder Development and Dairy Development.

The food of animal origin like milk, eggs and meat have shown a steady increase in production level during the last three decades. Though the department is a service department, revenue is being collected on milk by levy of cess on fluid milk coming from the neighbouring States, for distribution in Goa. The department has taken measures to control Anthrax and Foot and Mouth Diseases & Husbandry Services by resorting to vaccination of cattle. Goa is free from Rinderpest diseases and efforts to eradicate Foot and Mouth diseases are in progress.

Significant initiatives

- The infrastructure of hatchery for supply of day old chicks has been strengthened so as to increase the production of birds and eggs thereby meeting the demand of poultry farmers.
- An incentive of Rs. 2/- per lt. for milk production is in force. The incentive limit of 150 lts. for production of milk has been removed. The proposal has been approved and the implementation has already started during the current year.
- During the year 2007-08, till Nov.' 07, subsidy of Rs. 15,600/- has been released to one beneficiary for purchase of milking machine under 30% subsidy
- Medicines have been supplied to the various field establishments for mass deworming of cattle & buffaloes so as to cover maximum cattle and buffalo population to improve the health of the animals and increase productivity.
- Construction of milk booths at different places including tourist places and highways in order to make available milk federation products at Government rates during the day are proposed to be taken up by GSIDC very shortly. The

department has already proposed to take up 10 booths at first stage and the proposal is under process.

- Subsidy of Rs.200/- per tonne for transport of poultry feed from outside the State is being provided. During the year 2007-08 till November 2007, Rs. 1.26 lakh has been released as subsidy for transport of poultry feed from outside the State.
- Radio Frequency Identification Device is provided for cattle and buffalo covering about 15,000 animals. The beneficiaries who availed assistance under Kamdhenu scheme are being benefited under this scheme. Till date an amount of Rs. 2.73 lakh has been spent.

Achievements

- Veterinary Hospitals at Tonca, Panaji has been upgraded and modernized. State Disease Investigation Laboratory has been modernized with latest sophisticated equipment.
- Under the scheme poultry development, a modern poultry unit can also be set up of 2000 broilers 5000 layers for which 25% of the cost of the plant and machinery and equipment is granted as subsidy subject to a maximum of Rs. 2.00 lakh to an individual.
- 9 proposals for piggery development units have been received and same are under process. A subsidy of Rs. 49,412/- is given for starting a piggery unit of 20 sows and 2 boars and construction of sty to accommodate the animals.
- During the year 2007-08, 2 Modern Dairy units have been established till November 2007 and 4 units are under process.
- 7 seats have been reserved for students from Goa in Maharashtra Veterinary College and 5 seats in Pondichery Veterinary College. For the reservation of the seats a support fee is paid by the State Government @ Rs. 2.00 lakh per student for the full course. The scheme started in the year 1998-99 and till date 49 students have been deputed, of which 21 students have completed the course. During the current year, 5 students have been admitted.
- The department also assists the NGOs working in the field of animal welfare. There are four NGOs working in the field of animal birth control of stray dogs/cats and one NGO looking after the welfare of cattle which are old, wounded, and homeless. These NGOs are working for the implementation of the Prevention of Cruelty Act.
- The department collected revenue of Rs. 34.00 lakh till November 2007 and cess on fluid milk coming into the State amounting to Rs. 20.00 lakh.
- Liquid Nitrogen tanker has been purchased for supply of liquid nitrogen to every sub-centre on demand for preservation of frozen semen.
- The Government assists the milk producers by giving incentive on the milk produced and poured into the dairy society @ 10% upto 12% depending on the quantity of milk poured and the value

of the payment received by the farmers. During the year 2007-08, till date 4000 farmers have been benefited under the scheme.

- Under 50% subsidy for seasonal cultivation, 7 farmers have been assisted till November 2007 and subsidy amounting to Rs. 0.21 lakh has been sanctioned and 5 proposals are under process.

New initiatives

A new scheme for breeding of non-descript cows has been proposed with an objective

- To improve & sustain the productivity of non-descript livestock through use of artificial insemination practices.
- To compensate the farmers for the man-hours lost in approaching the local Veterinarian and for feeding the animals.
- To assist the farmer financially in stall feeding the non-descript animals.
- To reduce the menace of stray animals.
- It is proposed to increase working hours at various key establishments in the first phase.
- Artificial Insemination facilities will be provided at the door steps of the farmers.
- Extension services will be strengthened to rope in more entrepreneurs to take up animal husbandry activities. Giriraja birds, a mixture of hybrid and local birds have been introduced to provide nutritious protein requirement to small farmers and to encourage backyard poultry units. Hybrid pigglings of Yorkshire breed are proposed to be supplied to piggery farmers for cross breeding and upgrading local pig breed for increasing better efficiency and feed conversion among non-descript pigs.

3. FOREST DEPARTMENT

The most important function of the department is to protect the existing forest from all harmful agencies, biotic or abiotic. These are ensured through the enforcement of “The Indian Forest Act, 1927” and “The Forest Conservation Act, 1980”. Constant efforts are made to enhance the health and productivity of these forests. The blank and degraded areas of existing forests are restocked through annual reforestation programmes. Apart from protecting Government forest, the department is also responsible for Preservation of Tree cover even in private owned areas. The department strictly enforces provisions of the Preservation of Tree Act, 1984 and promotes planting of trees in private lands, Communidade lands, roadsides, etc. through a variety of packages including distribution of seedlings and development of decentralized People's nurseries. The talukawise distribution of areas under forest & forest produce are given in Annexure 20 & 21.

Wildlife management is another thrust area of the department. The department also promotes eco-tourism in those areas where

pristine wilderness beckons any nature lover. There is one National Park and six Wildlife Sanctuaries in the State, thus bringing about 62% of the forest area under Protected Area (PA) network. Regular patrolling of these areas is carried out and the Wildlife Protection Act, 1972 is vigorously and effectively enforced all over the State.

a. Initiatives taken to preserve forests

- Non invasive use of forest resources particularly of medicinal plants to be popularized.
- Promotion of Eco-tourism by bringing more potential areas within the ambit.
- Urban beautification and recreational forestry to be given special emphasis.
- Strict implementation of e-governance.
- Prioritisation of research to boost forest productivity and promote eco-friendliness.

b. Non-invasive use of Forest resources

Under the aegis of National Medicinal Board of Department of Ayurveda, Yoga and Naturopathy, Unani, Sidha and Homeopathy (AYUSH) plantations of medicinal plants is promoted as a value addition crop by the farmers. 31 species having good marketability have been identified which can be cultivated easily by the farmers in different regions of the country. For receiving grants under these schemes, the farmers/small entrepreneurs can formulate the schemes and submit the same through the Forest Department to National Medicinal Plant Board for seeking up to a maximum of Rs.10.00 lakh for pilot projects.

In order to create awareness on medicinal plants, demonstration plots of medicinal plants have been established at the Forest Training School Campus, Valpoi, Valkini, Sanguem. A model medicinal plant garden namely "Aushadhi Vatika" has also been created at Raj Bhavan, Dona Paula.

c. Greening India Programme

Ministry of Environment and Forests, Government of India have launched the Greening India Programme with a view to bring about 25% of the geographical cover of the country under vegetation by the year 2007 and 33% by the year 2012 respectively. This would be achievable only by bringing more areas under private community ownership and the waste land under the trees cover. Although the State of Goa is having 38% area under forest cover, any further addition to vegetation cover shall contribute to enhancement of national average. With a view to encourage afforestation activities on community/private lands, Government of India has launched Greening India Programme since the year 2005. Under this programme, production of quality planting material and tree planting activities are encouraged through the private agencies.

d. New initiatives

The department has taken initiative for the preparation of Master Plan of Bondla Zoo.

On completion and approval of the same, the entire Zoo will be re-designed and will be managed accordingly. A collection plan will be prepared detailing the species to be housed. Basic amenities for the visitors, a modern interpretation centre, etc, will also be developed.

For promotion of eco-tourism the department has already started services for ferrying the tourist to the Dr. Salim Ali Bird Sanctuary. A bus service from Campal to Bondla on weekend to be commenced shortly.

A camp site has been established near Collem entry gate of Bhagwan Mahavir National Park where tent accommodation has been developed to educate and sensitise the visitors towards conservation of nature. Protection of Wildlife Sanctuary and National Park has been ensured thereby bringing down the number of offence cases in the protected areas.

Achievements

- About 1,18,774 tourists of foreign and domestic origin visited the eco-tourist spots in the Wildlife Sanctuaries which covers about 62% forest area of the State. Eco-tourism guides have also been trained and authorized by the department, so that tours are conducted to interesting places and local employment is generated.
- The Municipal Garden, Margao is handed over to the Forest Department and the beautification work is being carried out on priority basis.
- The boundary of the protected areas has been consolidated near the vulnerable points to check poaching, illegal grazing and to reduce man animal conflict by means of constructing 1200 mts. length rubble wall (600 mts. each at Bhagwan Mahavir Wildlife Sanctuary and Cotigao Wildlife Sanctuary) and 1.75 kms. Cattle Proof Trench (1000 mts. at Cotigao Wildlife Sanctuary and 750 mts. Bhagwan Mahavir Sanctuary and National Park). 40 check dams have been constructed for soil and water conservation. An area of 6 ha. of fruits bearing trees have been planted to supplement the availability of fruit for the wild animals. 6 ha. of grassplots have been maintained for the herbivorous. 45 water holes are maintained to avoid scarcity of water for wild animals.
- The Wildlife Rescue Squad of the department operating from Campal, conducts rescue operation for wildlife animals, which have strayed into human habitations. These animals are likely to be killed or mauled seriously, unless they are attended on priority and hence the squad has to be on duty round the clock. In many cases, the animals which require medical attention are administered in the Rescue Centers. It is always ensured that only those animals, which have restored their health completely, are left back into the wild.
- During the year 2007-08, 1071 reptiles, 47 mammals, 14 birds and 2 other wild

animals were rescued.

e. Goa Forest Development Corporation

The Goa Forest Development Corporation has been implementing the Government. of India scheme under the Joint Forest Management on Care and Share basis, by registering the local Self Help Group with the Corporation for management, maintenance and fresh planting areas, so as to involve rural people in protection management and development of cashew plantation and to provide job avenues for them to supplement their family income thereby lifting their socio-economic status. The Corporation plays a lead role in the protection and conservation of resources with due regards to the environment and ecology.

f. Significant achievements

- During the last 3 years, replanting of refractory areas have been taken up under the schemes of National Horticulture Mission (NHM) by planting high yielding variety of cashew grafts Vengurla-4 (V4),Vengurla-7 (V7) which will shortly go into production. Also during the year, the implementation of Frontline Technology of NHM in the talukas of Ponda & Sattari as demonstration plots with high yielding variety of cashew grafts has been taken up.
- The pilot project on Apiary has been implemented during the current year. To supplement income to the Corporation, Beekeeping activities in the Rubber plantations at Shishewal in Canacona and Veluz in Valpoi has been started. There are also plans of commercial production and registration under the trade mark of AGMARK which is in the process.
- Establishment of the prestigious "Botanical Garden Cum Eco-Recreational Park" project situated on the downstream side of the Salaulim Dam is a major notable achievement of the Corporation. The objective of the project is to promote eco-tourism in the hinterland area of South Goa. The work on the 1st Phase of the project was carried out and the Irrigation Sprinkler System has been installed for watering the garden. The implementation of the major component works is under process.
- Auction of fruit bearing areas as per time schedule and also maintenance of nurseries at Kankire, Dhavali (Ponda), Veluz (Valpoi) and Fondsorem (Canacona) for new stock of grafts as well as seedlings has been taken up.
- Implementation of the Front Line Technology in the areas of Ponda & Sattari Talukas.
- Maintenance, management and upkeep of Rubber plantations and also motivation for rubber tappers/ processors to help boost productivity by providing proper tools/ implements such as searchlight, gumboots, knives, raincoats, etc. is being taken regularly.

- The Range Forest Officer of Cancona have been bestowed with a State award for best tree plantation.

4. FISHERIES

The fishing industry in the State mainly comprises of mechanized vessels, fish processing plants, marketing network and shrimps farms which are mainly in the private sector. The State Government has been developing necessary infrastructure required for the industry. There is scope for expansion and upgradation of the existing fishing fleet in the private sector. Government of India is also encouraging construction of deep sea fishing vessels upto the cost of Rs 40.00 lakh with back-handed subsidy of Rs.15.00 lakh and is seeking private investment for the same. Further, financial assistance is also available to private sector for construction of fish processing plants upto a limit of 25 % for installation of cold storage and ice plants, purchase of refrigerated vans etc. The State Government is propogating the schemes to seek private investment for such units. The general industrial climate in the State as regards to fishing industry is encouraging. It is proposed to upgrade the major landing centres, two each in North and South Goa districts. A fish meal plant is envisaged to be established at an appropriate site in South Goa.

Satisfactory service is being rendered by the department to the general public and fishermen in particular. The department has already supplied adequate number of Citizen Charters on Fisheries to the local bodies like Village Panchayats as also to various Government and Non Government Organization. Considering the increased scope of work of the department and inorder to facilitate the fishermen of South Goa district, it has been decided to open a new office in South Goa.

The department has made significant development in the existing infrastructure by providing adequate landing and berthing facilities to the traditional fishermen and fishing boat operators.

a. Various infrastructure works undertaken/in progress during the year 2007-08

- Construction of fishing jetty phase III of 50 mts at Cutbona
- Widening road at Cutbona jetty
- Netmending shed at Nerul and extension of fish landing ramp at Marvel Dona-Paula
- Compound wall at Ela Dhauji farm
- Compound wall for the proposed aquarium site at Keri- Sattari.

b. Achievements under various schemes

- Financial assistance for purchase of InBoard/OutBoard Motors to the extent of 50% of the cost limited to Rs.25,000/- per OBM will assist 50 beneficiaries during 2007-08.
- The beneficiary is entitled for subsidy of

Rs. 8000/- for purchase of seagill nets. 50 beneficiaries shall be assisted during 2007-08.

- 97 fishermen were assisted during 2006-07 and 50 beneficiaries are proposed to be assisted during 2007-08 with assistance for construction/ purchase of new wooden FRP crafts. The subsidy is to the extent of 50% of the cost limited to Rs.40,000/-.
- Under Group Accident Insurance Scheme, 3113 fishermen were insured during the year 2006-07. The dependent of the insured fishermen will receive an amount of Rs.50,000/- in the event of death and Rs.25,000/- on partial disability.
- Under the scheme Stipends for various fisheries training courses, the trainee (fishermen) will be eligible for a stipend of Rs.500/- p. m., Rs.20/- as food allowance per day during practical training and Rs.50/- as uniform allowances per month. 15 trainees have been enrolled during 2007-08.
- Under the scheme Financial assistance for purchase of Insulated Boxes to fisher persons engaged in selling and marketing of fish and fishing at sea., a beneficiary is entitled for grant of Rs.3000/- and Rs.4000/- respectively. 645 Insulated Boxes have been given to the fishermen during 2006-07 and it is expected that 500 beneficiaries will be covered during 2007-08.
- 20 ha. area has been developed during 2006-07 under the scheme Financial assistance for development of shrimp farms @ Rs.30,000/- per ha. The beneficiary is entitled for a maximum of Rs.3.00 lakh i.e. upto maximum area of 10 ha.
- Under the Centrally Sponsored Scheme "Fishermen Development Rebate" on HSD oil @ Rs.1.50 per litre consumed by fishing vessels below 20 mts. length was granted to 773 beneficiaries during 2006-07.
- With a view to give relief to the traditional fishermen who are compelled to buy kerosene at high rates from open market, a special quota of kerosene has been requisitioned from the Government of India and the subsidy is released to OBM operators towards purchase of kerosene per year on maximum consumption of 2000 lts. of kerosene. 564 fishermen were given subsidy during 2006-07 and 750 fishermen will be covered during 2007-08.
- Under General Insurance Scheme for fishermen, 9 accidental claims were settled during the year 2006-07 and Insurance cover to 5000 fishermen will be given during 2007-08.
- 69 cases were settled in 2006-07 and 350 cases will be settled during 2007-08 under the Saving cum Relief scheme.

c. Fish Catch and export

- Annual fish catch in Goa during the years 2000 to 2006 is given in table 4.3. The trend in export of marine products during 2000-2007 is given in table 4.4.

Table 4.3 : Trend in Fish Catch 2000-2006

Sl. No	Year	Annual Fish Catch in tonne		
		Marine	Inland	Total
1	2000	64563	3570	68133
2	2001	69386	3749	73135
3	2002	67563	3684	71247
4	2003	83756	4283	88039
5	2004	89932	4396	94328
6	2005	103091	4196	107287
7	2006	98643	3903	102546

Table 4.4 : Export of Marine Products 2000-2007

Sl. No	Export of Marine Products		
	Year	Quantity (in tonne)	Value (Rs. in lakh)
1	2000	10732	3357
2	2001	7714	3007
3	2002	15594	5620
4	2003	10288	3273
5	2004	8856	3909
6	2005	11001	6178
7	2006	8472	6438
8	2007	11001	7300

5. COOPERATION

a. The Goa Co-operative Societies Act, 2001

So far, the Co-operative movement in the State is governed under the Maharashtra Co-operative Societies Act, 1960. Now, the State has developed its own Act and the Legislative Assembly has passed the Goa Co-operative Societies Act, 2001 on 29-3-2001 and was assented by the Governor of Goa on 18/05/2001. Necessary Rules under the Act have been framed and published in the Official Gazette. The new Act encompasses unique features like curtailment of arbitrary powers of the Registrar, more autonomy to co-operative movements, removal of quasi-judicial powers of the Registrar, formation of representative general body for societies with large membership, establishment of Co-operative Authority for deciding disputes etc. The New Act and Rules will be implemented once the required infrastructure gets underway.

b. Co-operative Audit

The co-operative audit system has been streamlined by evolving guidelines and norms for awarding Audit Classifications so as to grade performance of societies in a rational and impartial manner. In order to ensure orderly functioning of Urban Credit Co-operatives and to avoid indiscreet competition in mobilizing deposits, advancing loan etc., by them, necessary instruction/norms have been laid down to have uniformity in interest rate on deposits/loans. Maintaining required ratios between resources and advances have also been laid down.

c. Significant initiative for Development of Co-operative sector in the State

The co-operative sector in the State of Goa plays a vital role in providing support to key sectors like agriculture, small scale industry etc. which has helped to create employment opportunities to a great extent. However, on account of unhealthy practices adopted by the Management of some prominent institutions and due to global phenomena of recession prevailing in the economic sector during the last few years, the financial position of some co-operative institutions which had a good track record in the past, has weakened thereby resulting in deterioration of investor's faith in these institutions. These institutions have spread their operations at the grass root level all over the State and possess inherited strength. Keeping this in view, the Government has launched various remedial initiatives including financial support in order to put these institutions back on the right track.

- The Sanjivani Sahakari Sakhar Karkhana is the only Sugar Factory in the State. With a view to protect the interest of cane growers in the State and to enable the Karkhana to overcome the financial difficulties, the State Government provides financial support to the Karkhana by way of share capital contribution. To achieve maximum possible crushing and to keep the losses to minimum level the factory imports sugarcane from the neighboring States like Maharashtra and Karnataka. The efforts are being made to increase the local sugarcane production. Revamping of the factory is also being explored. The Sugar Factory has decided to increase its sugar production, which will enable it to mobilize funds to undertake modernization of the old and out dated machinery for improving its efficiency. During the crushing season 2006-07, the factory has crushed 200869.631 M.T. of sugar cane and produced 19268.50 quintals of sugar. During the year 2006-07, the Karkhana have been provided financial support of Rs.3.00 crores by way of share capital contribution. As on 31-03-2007 the total amount of Govt. investment by way of share capital contribution to the Sugar Factory is Rs.30.00 crore and Rs.7.75 crore by way of loan.
- It is proposed to provide financial

support by way of loan of Rs.1.00 crore to the Goa State Co-operative Marketing and Supply Federation to enable the Federation to pay off high cost loans availed from the Goa State Co-operative Bank and to restore its business activities, which are stagnated at present due to dearth of financial resources. The Government has also impressed upon the Management to implement strict economic measures to bring down the managerial cost to the possible minimum level.

- It is also proposed to strengthen the share capital base of the Goa State Co-operative Bank and the Goa State Co-operative Housing Finance & Federation Ltd., Panaji to enable them to raise the required finance for enhancing their business activities.

d. Primary Agricultural Cooperative Societies

The recommendations made in the TASK FORCE REPORT submitted by Vaidhyanathan Committee for revival of rural cooperative credit institutions have been accepted by the State Government and the same is vetted by the Law Department. To begin with, it is proposed to provide assistance to the Primary Agricultural Credit Cooperative Societies for cleansing of their balance sheets, which are impaired due to non-recovery of loans so that they can rejuvenate their activities. Certain prerequisites prescribed in the recommendation made by Vaidhyanathan Committee have already been incorporated in the New Cooperative Societies Act 2001. As such Government has not signed the MOU with the Central Government.

e. Agricultural Marketing

The Goa State Agricultural Marketing Board (formerly known as Goa Agricultural Produce Market Committee) has been constituted and entrusted with the task of establishment of market yards in all the important talukas of the State. So far eight market yards have been established. These yards provide with infrastructural facilities for the better marketing of the agricultural produce.

With the aim to replace Maharashtra Agricultural Produce Marketing (Regulation) Act, 1963 which is in force in the State, the Department of Cooperation have already enacted a Legislation called "The Goa Agricultural Produce (Development and Regulation) Act, 2007" and all the reformatory provisions suggested by the Government of India in its Model Act which provides for direct purchase centres, consumer/farmers markets for direct sale and promotion of Public Private Partnership in the management and development of agricultural markets including the provisions on Contract Farming have been incorporated therein. This Act has been passed by the Legislative Assembly on 8-3-2007 and assented by the Governor on 1-08-2007. The same has been published in the Official Gazette vide Series I No.18 dated 6-08-2007. As regards rules, the same are drafted and

are being forwarded to the Government for approval.

Achievements

- The Goa State Co-operative Milk Producers Union Ltd, Curti, Ponda has procured 164.33 lakh litre of milk.
- 69 New Co-operative Societies have been registered.
- The department has completed the task of feeding of data required under Right to Information Act, 2005 and the work of feeding of data on the total organization set up of the department required for e-governance is underway.
- The Goa Bagaytdar Sah. Kharedi Vikri Saunstha Maryadit, Ponda, Mandovi Fishermen Marketing Coop. Society Ltd., Malim and Coutband Fisheries Cooperative Society Ltd. Velim, Salcete has been conferred upon by the NCDC, New Delhi for their outstanding performance in the Coop. sector of this State.